

Vanya Peter Cohen

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I co-created OpenGPT-2, the first open-source replication of OpenAI's GPT-2 and OpenWebText, a widely used dataset for LLM training and evaluation. My PhD research focuses on enabling agents to understand and act in complex environments, spanning grounded language understanding, reinforcement learning, language-guided robotics, and VLM reasoning, with papers at ICML, EMNLP, IJCAI, ICRA, and IROS.

Education

- 2021-Now **Ph.D. in Computer Science**, The University of Texas at Austin.
 - Advised by [Ray Mooney](#). Expected Graduation: 08/26.
- 2017-2018 **M.Sc. in Computer Science**, Brown University, Providence, RI.
 - Co-advised by [Stefanie Tellex](#) and [George Konidaris](#).
- 2013-2017 **B.Sc. in Computer Science**, Brown University, Providence, RI.

Research Experience

- 2025-Now **Google DeepMind**
Student Researcher.
Focus Areas: Reinforcement learning and computer-control agents.
- 2021-Now **Machine Learning Research Group - UT Austin**
Ph.D. Student.
Focus Areas: Grounded natural language processing, reinforcement learning, robotics; Language-driven agents interacting in complex environments.
- 2020-2025 **Blackbird.AI**
AI Scientist '24 & Consultant/ML Engineer '20-'23.
Focus Areas: Multimodal agents for fact-checking; dataset creation and training pipelines for custom ML models.
- 2019-2020 **Luminoso Technologies Inc.**
Machine Learning Engineer.
Focus Areas: ConceptNet; Language models for semantic networks.
- 2017-2019 **Humans to Robots Lab – Brown University**
Research Assistant; Advised by Stefanie Tellex and George Konidaris.
Focus Areas: Language instruction following for manipulation, human-robot interaction, language grounding.
- 2016-2017 **Brown University Graphics Lab**
Undergraduate Research Assistant; Advised by [Andy van Dam](#).
Focus Areas: Collaborative gesture & multi-touch user interfaces.

Honors & Awards

- 2023 **Phi Kappa Phi Honor Society**
Inducted for academic excellence during Ph.D. program.

Peer-Reviewed Academic Publications

- ICML 2026 **MET-Bench: Multimodal Entity Tracking for Evaluating the Limitations of Vision-Language and Reasoning Models**
Vanya Cohen, Raymond Mooney
International Conference on Machine Learning (ICML), 2026
- RLC 2025 **Compositional Instruction Following with Language Models and Reinforcement Learning**
TMLR 2024 Vanya Cohen*, Geraud Nangue Tasse*, Nakul Gopalan, Steven James, Matthew Gombolay, Ray Mooney, Benjamin Rosman
Reinforcement Learning Conference (RLC) 2025; Transactions on Machine Learning Research (TMLR), 2024
- EMNLP 2024 **CaT-Bench: Benchmarking Language Model Understanding of Causal and Temporal Dependencies in Plans**
Yash Kumar Lal*, Vanya Cohen*, Nathanael Chambers, Niranjan Balasubramanian, Raymond Mooney
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2024
- IJCAI Survey Track 2024 **A Survey of Robotic Language Grounding: Tradeoffs Between Symbols and Embeddings**
Vanya Cohen*, Jason Xinyu Liu*, Raymond Mooney*, Stefanie Tellex*, David Watkins*
International Joint Conference on Artificial Intelligence: Survey Track (IJCAI: Survey Track), 2024
- ICRA 2024 **CAPE: Corrective Actions from Precondition Errors using Large Language Models**
Shreyas Sundara Raman, Vanya Cohen, Ifrah Idrees, Eric Rosen, Ray Mooney, Stefanie Tellex, David Paulius
IEEE International Conference on Robotics and Automation (ICRA), 2024
- IROS 2019 **Grounding Language Attributes to Objects using Bayesian Eigenobjects**
Vanya Cohen*, Benjamin Burchfiel*, Thao Nguyen*, Nakul Gopalan, Stefanie Tellex, George Konidaris
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2019

Non-Archival Publications & Preprints

- NeurIPS 2025 **MET-Bench: Multimodal Entity Tracking for Evaluating the Limitations of Vision-Language and Reasoning Models**
Workshop Vanya Cohen, Raymond Mooney
Multimodal Algorithmic Reasoning at NeurIPS, 2025
- ICML 2025 **MET-Bench: Multimodal Entity Tracking for Evaluating the Limitations of Vision-Language and Reasoning Models**
Workshop Vanya Cohen, Raymond Mooney
Workshop on Assessing World Models at ICML, 2025
- ACL 2023 **Using Planning to Improve Semantic Parsing of Instructional Texts**
Workshop Vanya Cohen, Raymond Mooney
Workshop on Natural Language Reasoning and Structured Explanations at ACL, 2023

- CoRL 2022 **End-to-End Learning to Follow Language Instructions with Compositional Policies**
Workshop **Vanya Cohen***, Geraud Nangue Tasse*, Nakul Gopalan, Steven James, Ray Mooney, Benjamin Rosman
Workshop on Language and Robotics at the Conference on Robot Learning (CoRL), 2022
- NeurIPS 2022 **Planning with Large Language Models via Corrective Re-Prompting**
Workshop Shreyas Sundara Raman, **Vanya Cohen**, Eric Rosen, Ifrah Idrees, David Paulius, Stefanie Tellex
Foundation Models for Decision Making Workshop at NeurIPS, 2022
- AAAI-FSS 2021 **Learning to Follow Language Instructions with Compositional Policies**
Vanya Cohen*, Geraud Nangue Tasse*, Nakul Gopalan, Steven James, Matthew Gombolay, Benjamin Rosman
AI-HRI Symposium at AAAI Fall Symposium Series (AAAI-FSS), 2021
- NeurIPS 2019 **OpenGPT-2: Replicating a 1.5 Billion Parameter Language Model**
Workshop Aaron Gokaslan*, **Vanya Cohen***, Ellie Pavlick, Stefanie Tellex
NeurIPS NewInML Workshop, 2019

Invited Articles

- XRDS 2020 **OpenGPT-2: Open Language Models and Implications of Generated Text**
Vanya Cohen, Aaron Gokaslan
XRDS: Crossroads, The ACM Magazine for Students, Fall 2020

Speaking & Interviews

- Hack@Brown 2024 **Brown University Hackathon Alumni Panelist**
Invited speaker for the alumni panel at Brown's annual hackathon.
- Wired August 2019 **OpenAI Said Its Code Was Risky. Two Grads Re-Created It Anyway.**
Interviewed by Tom Simonite.

Other Work Experience

- 2017-2018 **National Bureau of Economic Research - UCLA/Brown**
Part-time Research Engineer.
Developed AWS web-scrappers and parsers to construct large-scale datasets of financial disclosures for an econometrics research team.
- Summer 2016 **TripAdvisor**, Palo Alto, CA
Software Engineering Intern (Full-stack iOS Developer).
Optimized iOS map features, cutting load times by 50%, and redesigned travel guides to boost user engagement.

Professional Activities

Workshop Organization

- NeurIPS 2020 **Co-Organizer for NewInML: A Workshop for Newcomers to Machine Learning at NeurIPS 2020**
Workshop Zhen Xu (UPSAclay), **Vanya Cohen**, Shruti Mishra (Harvard), MingYu Lu (UW)

Reviewer

- 2026- Neural Information Processing Systems (NeurIPS)
- 2026- International Conference on Machine Learning (ICML)
- 2025 Association for Computational Linguistics (ACL) Rolling Review (ARR)
- 2025- International Joint Conference on Artificial Intelligence: Survey Track (IJCAI: Survey Track)
- 2024 Robotics: Science and Systems (RSS)
- 2023 International Conference on Robotics and Automation (ICRA)

University Service

- 2022-2024 Organized the Connecting Language and Machine Perception (CLAMP) Reading Group, bringing together students and faculty from NLP, robotics, CV, linguistics, and CogSci to discuss recent grounded language processing papers.

Skills: Python, PyTorch, TensorFlow, HuggingFace, C++, AWS, GCP, MTurk/crowdsourcing.